



Indian Institute of Technology Ropar
Department of Mathematics
MA302 - Optimization Methods with Applications
Second Semester of the Academic Year 2025-26

Tutorial sheet - 6

1. Write the dual of the following (primal) LPP and solve the dual by solving the primal

$$\text{Minimize } Z = 5x_1 + 3x_2$$

subject to

$$2x_1 + 4x_2 \leq 12,$$

$$2x_1 + 2x_2 = 10,$$

$$5x_1 + 2x_2 \geq 10,$$

$$x_1, x_2 \geq 0.$$

2. Consider the following LP problem:

$$\text{Maximize } Z = 3x_1 + 2x_2 + 5x_3$$

subject to

$$x_1 + 2x_2 + x_3 \leq a_1,$$

$$3x_1 + 2x_3 \leq a_2,$$

$$x_1 + 4x_2 \leq a_3,$$

where a_1 , a_2 and a_3 are constants.

For specific values of a_1 , a_2 and a_3 the optimal solution is given by the following tableau:

Basic	x_1	x_2	x_3	x_4	x_5	x_6	Solution
z	4	0	0	c_1	c_2	0	1350
x_2	b_1	1	0	$\frac{1}{2}$	$-\frac{1}{4}$	0	100
x_3	b_2	0	1	0	$\frac{1}{2}$	0	c_3
x_6	b_3	0	0	-2	1	1	20

where b_i 's and c_i 's are constants.

Determine:

- (i) the values of a_1 , a_2 and a_3 that yield the given optimal solution.
(ii) the values of b_1 , b_2 , b_3 and c_1 , c_2 , c_3 in the optimal tableau.
3. Solve the following LPP by the dual simplex method

$$\textbf{Minimize} \quad Z = x_1 + 2x_2 + 3x_3 + 4x_4$$

subject to

$$\begin{aligned}x_1 + 2x_2 + 2x_3 + 3x_4 &\geq 30, \\2x_1 + x_2 + 3x_3 + 2x_4 &\geq 20, \\x_1, x_2, x_3, x_4 &\geq 0.\end{aligned}$$

4. Use the dual simplex method to check that the following LPP is infeasible.

$$\textbf{Maximize} \quad -x_1$$

subject to

$$\begin{aligned}x_1 - x_2 &\geq 3, \\-x_1 + x_2 &\geq 4, \\x_1, x_2 &\geq 0.\end{aligned}$$